Spectrum-Conditioned Relative Reranking for MS/MS Molecule Retrieval

Abstract. Retrieving molecular structures from tandem mass spectra is difficult because mass- and formula-constrained candidate pools contain chemically similar hard negatives. We study a direct, non-generative second stage that learns to compare those candidates using the query spectrum rather than relying only on a spectrum–molecule cosine score. SpecEmbedding-Rerank first aligns a peak-sequence spectrum encoder with a molecular graph encoder and caches the complete supplied pool (at most 256 candidates). It then performs pointwise coarse scoring followed by a spectrum-conditioned relative candidate module on the coarse top 40. The relative preference is constructed as an antisymmetric pair difference and is equivariant to candidate-list permutation; no retrieval-rank feature or training-time positive insertion is used. We compare the relative model with a capacity-matched no-rank pointwise control and retain the earlier candidate-set Transformer only as a historical secondary baseline. The main evaluation uses the MassSpecGym 1.3.1 retrieval candidate order and its two-dimensional InChIKey identity rule, with a tracked full-pool audit establishing one identity-equivalent positive per query in the saved pools. A local exact-target-SMILES table is retained as a sensitivity view. Candidate coverage, seed units, saved-embedding scope, and incomplete comparisons are stated alongside the results. The evidence is intended to establish whether explicit spectrum-conditioned candidate relations add value beyond coarse supervised reranking, not to claim a universal self-attention or external-baseline gain.

Keywords: Data mining for bioinformatics · Tandem mass spectrometry · Molecule retrieval · Learning to rank · Multimodal representation learning

1 Introduction

MS/MS is widely used to identify small molecules in metabolomics, natural-product discovery, and related biomedical applications. An observed spectrum is only a partial, condition-dependent view of a molecular structure: different molecules can yield similar fragments, while the same molecule can produce different spectra across instruments, adducts, and collision energies. Consequently, structure annotation is naturally a large-scale data mining problem in which a query spectrum must retrieve and rank a correct structure from a chemically constrained candidate pool.

MassSpecGym provides structure-disjoint data splits and standardized mass- and formula-conditioned candidate pools for this problem [3]. Recent methods mainly address the cross-modal gap between spectra and molecules. JESTR learns a spectrum–molecule joint space with contrastive multiview coding and ranks candidates directly by cosine similarity [12,16]. GLMR instead retrieves contextual candidates, generates a refined molecule with a conditional language model, and reranks candidates by molecule–molecule similarity [20].

Both approaches leave an important data mining question open. Candidate-wise cosine scoring does not explicitly train a second-stage objective over the retrieved set, while generative retrieval adds an autoregressive decoder and makes ranking depend on the quality and validity of a generated structure. We ask whether a supervised second-stage ranker can separate hard candidates and whether a spectrum-conditioned relative comparison adds value beyond a capacity-matched pointwise scorer. The question is deliberately narrower than claiming a new universal retrieval model: the candidate pool, identity rule, and first-stage checkpoint remain fixed while the second-stage mechanism is audited.

Building on the peak-sequence spectrum architecture introduced by SpecEmbedding [18], we construct SpecEmbedding-Rerank, a two-stage retrieval and reranking framework. The first stage trains that spectrum tower from scratch and aligns it with a GINE molecular graph encoder. The second stage scores the full supplied pool with an absolute branch, then refines only a coarse top-K_r subset with spectrum-conditioned candidate relations. Pair preferences are formed by subtracting the score of the reversed ordered pair, which gives antisymmetry without a rank embedding. Training uses only naturally present positives; the implementation never repairs a truncated list by inserting its target. A pointwise model with the same absolute branch provides the primary capacity-controlled comparison, while the earlier candidate-set Transformer is kept as a secondary historical baseline.

Our contributions are:

- We instantiate and audit direct, non-generative second-stage learning to rank on MassSpecGym candidate pools, with full-pool coarse scoring followed by spectrum-conditioned relative candidate comparison.
- We make the mechanism testable: the relative branch is rank-free, antisymmetric, permutation-equivariant, and trained without positive forcing; a capacity-matched pointwise control isolates the contribution of candidate relations.
- We report a three-seed MassSpecGym 1.3.1-compatible candidate-order and two-dimensional-identity evaluation using saved embeddings, retain a local exact-SMILES sensitivity view, and separate the new method from the historical Transformer audit. Full end-to-end loader re-encoding, external baselines, and deployment efficiency remain explicit limitations rather than being filled with uncontrolled comparisons.

AI-assistance disclosure. OpenAI Codex assisted throughout all sections with drafting and language revision, and during the project with code editing, experiment orchestration, and consistency audits. Reported observations and numerical results came from the stated software pipeline, not model generation. The authors reviewed all AI-assisted text and code, verified the technical claims, citations, saved artifacts, and reported results, and assume full responsibility for the manuscript's accuracy, completeness, and originality.

2 Related Work

2.1 Annotation Paradigms

Computational MS/MS annotation differs in retrieved objects and ranking evidence. Candidate-score correction predates our work: MetFusion combines in-silico fragmentation with spectral-library evidence [8]; MS2Query reranks its top 2,000 MS2DeepScore matches with a five-feature random forest [11]; and LC-MS2Struct combines an MS/MS scorer with retention order to jointly rank candidates from one LC analysis [1]. We instead rank structures in query-specific MassSpecGym pools from frozen spectrum–molecule representations, without spectral-library-neighbor or retention-time evidence [3].

Within structure-database retrieval, CSI:FingerID and MIST predict molecular fingerprints from spectra [7,9]. CMSSP and JESTR learn joint spectrum–molecule spaces; MVP extends contrastive alignment across molecular graphs, fingerprints, individual spectra, and consensus spectra [5,12,6]. These methods order candidates by predicted-fingerprint scores or cross-modal similarity. SpecEmbedding instead studies spectrum–spectrum representation learning [18]; we reuse only its peak-sequence architecture, train it from scratch against molecular graphs, and learn a complementary second-stage order within a retrieved hard-candidate list.

Other paradigms use different intermediate objects. MassFormer predicts tandem spectra from molecular graphs [19], while MSNovelist generates structures from MS/MS-derived fingerprints [15]. GLMR conditions its molecular decoder on retrieved candidates and reranks them by similarity to the generated molecule [20]. Our closed-set method does not decode a structure or return a molecule outside the first-stage list; it directly optimizes the scores of available candidates.

2.2 Learning to Rank Candidate Sets

Learning-to-rank objectives define supervision over preferences or complete lists rather than treating similarity sorting as a fixed post-processing step. RankNet learns pairwise preferences, whereas ListNet defines a probability model over a ranked list [2,4]. Ranking functions can also exploit relations among competing items. Set Transformer provides attention blocks for set-structured inputs [13], while SetRank uses self-attention to model cross-document interactions [14]. We apply these principles to cached cross-modal embeddings with explicit spectrum–candidate pair features and first-stage score and rank priors. Table 1 positions the resulting query-specific, non-generative second stage among selected related MassSpecGym approaches. The contribution is neither reranking nor self-attention itself; residual scoring and candidate interaction are evaluated design choices rather than assumed sources of gain.

Table 1. Mechanistic comparison with selected related MassSpecGym approaches.

Model	Ranking signal	Candidate interaction	Generates	Training stages
JESTR	Spectrum–candidate cosine	None at inference	No	InfoNCE; optional late-stage candidate regularization
GLMR	Generated–candidate cosine	Retrieved candidates condition the generator	Yes	Dual-path InfoNCE plus generation
Ours	Residual score from pair features and retrieval priors	Optional candidate self-attention	No	Cross-modal InfoNCE plus listwise/pairwise reranking

3 Problem Formulation

Let s be an MS/MS spectrum, m^+ its correct molecule, and $\mathcal{D}_s$ a query-specific mass- or formula-conditioned candidate pool. A base retriever assigns

$$b(s, m) = \mathrm{sim}(f_s(s), f_m(m)) \tag{1}$$

to every $m \in \mathcal{D}_s$ and returns the top-K list $\mathcal{C}_s = (m_1, \dots, m_K)$. The reranker learns scores $g(s, m_i, \mathcal{C}_s)$ such that m^+ is ordered as high as possible. Because it cannot return a molecule outside $\mathcal{C}_s$, end-to-end Recall@k is upper-bounded by

$$U_K = \frac{1}{N} \sum_{n=1}^{N} \mathbb{I}[m_n^+ \in \mathcal{C}_{s_n}]. \tag{2}$$

Define the within-list top-k accuracy conditional on first-stage coverage as

$$A_{k|K} = \frac{\sum_{n=1}^{N} \mathbb{I}[m_n^+ \in \mathcal{C}_{s_n},\, \mathrm{rank}_g(m_n^+) \leq k]}{\sum_{n=1}^{N} \mathbb{I}[m_n^+ \in \mathcal{C}_{s_n}]}, \qquad \text{Recall@}k = U_K A_{k|K} \leq U_K. \tag{3}$$

A reranker applied to a fixed cache can therefore change only $A_{k|K}$, not U_K. We report metrics over all queries, assigning zero credit when the positive is absent rather than conditioning the primary metric on successful first-stage retrieval.

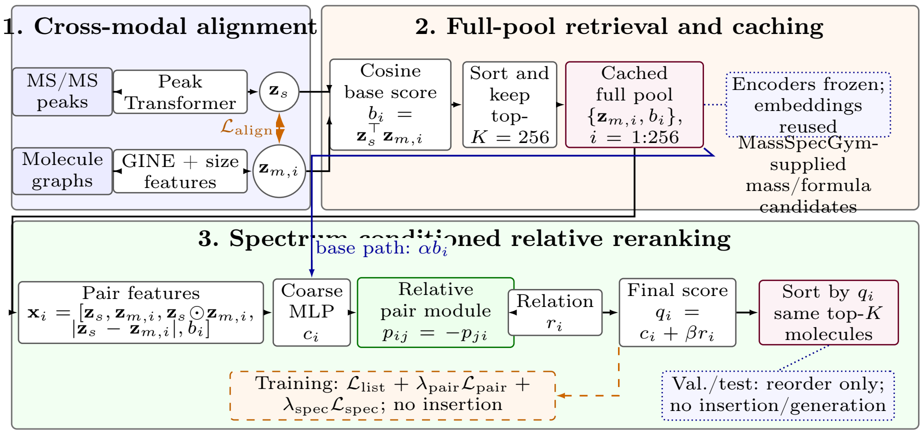

Fig. 1. Overview of SpecEmbedding-Rerank. Panel 1 aligns the spectrum Transformer and molecular GINE. Panel 2 uses their frozen, cached embeddings to produce the full candidate pool. Panel 3 applies pointwise coarse scoring followed by spectrum-conditioned relative candidate preferences on the coarse top-40; the legacy Transformer is a secondary baseline. Dashed arrows denote training-only operations. Positives are never forced into a truncated list, and no molecule is generated.

4 Method

4.1 Overview

The framework has three steps: cross-modal alignment, full-pool retrieval, and supervised coarse-to-fine relative reranking. Spectrum and molecule encoders are frozen while the reranker is trained, so their embeddings can be computed once and reused.

4.2 Cross-Modal Base Retriever

The spectrum encoder follows SpecEmbedding's peak-sequence architecture [18], which processes peak m/z values and intensities with sinusoidal m/z features and a Transformer. In our implementation, each spectrum has at most 100 tokens: a precursor-m/z token with intensity 2, followed by up to 99 fragments selected by intensity. Selected fragments retain their source-array order, and their intensities are normalized by the largest selected fragment intensity. Only these m/z–intensity tokens enter the spectrum encoder; adduct, instrument, and collision-energy metadata are not explicit inputs. We initialize this tower from scratch and train it against molecules rather than spectra. The molecular encoder is a GINE network over atom and bond features [10]. Mean-pooled graph features are augmented with explicit graph-size features. Two projection heads map the encoders to a common d-dimensional space:

$$\mathbf{z}_s = \frac{p_s(f_s(s))}{\|p_s(f_s(s))\|_2}, \qquad \mathbf{z}_m = \frac{p_m(f_m(m))}{\|p_m(f_m(m))\|_2}. \tag{4}$$

We use a bidirectional cross-modal contrastive loss that permits multiple positives. Spectra and molecules sharing a molecular label form positives, which avoids treating replicate spectra as false negatives. For a minibatch of size B, let $P(i)$ be the indices sharing the label of item i. With a learned inverse-temperature scale a, the base logits are

$$\ell_{ij} = a\, \mathbf{z}_{s_i}^\top \mathbf{z}_{m_j}. \tag{5}$$

The spectrum-to-molecule term is

$$\mathcal{L}_{s\to m} = -\frac{1}{B}\sum_{i=1}^{B} \frac{1}{|P(i)|} \sum_{j\in P(i)} \log \frac{\exp(\ell_{ij})}{\sum_{k=1}^{B}\exp(\ell_{ik})}, \tag{6}$$

and the molecule-to-spectrum term applies the same expression to $\ell^\top$. The alignment loss is their mean. The implementation parameterizes $a = \exp(u)$ and caps it at 100. At inference, the normalized dot product $b_i = \mathbf{z}_s^\top \mathbf{z}_{m_i}$ is the base score.

4.3 Candidate Construction and Training Protocol

MassSpecGym supplies target-molecule-keyed candidate lists; each query reuses the list keyed by its target molecule. The base retriever scores the complete pool of at most 256 molecules. The new coarse-to-fine reranker first scores this full pool pointwise and then applies a relation module to the coarse top $K_r = 40$ candidates. Both 256 and 40 are fixed design and compute-budget choices for this revision; the evaluation script can truncate a saved pool without inserting a positive target.

Training uses only queries whose target naturally occurs in the full 256-candidate pool. Queries without a natural positive remain in coverage statistics but do not enter the reranker loss. Candidate order is shuffled during training while each candidate's embedding and base score move together. No rank embedding is used anywhere in the new model, and the cache-compatible `base_ranks` field is ignored by its forward path. In particular, no positive is inserted into a truncated list. The target is present in all 32,010 target-keyed source lists, so the full-pool cache does not enlarge the supplied candidate set; missing positives arise only from first-stage coverage when a smaller evaluation cutoff is requested.

4.4 Spectrum-Conditioned Relative Candidate Reranker

For candidate m_i we retain an absolute compatibility vector, but omit the retrieval rank:

$$\mathbf{x}_i = [\mathbf{z}_s,\ \mathbf{z}_{m_i},\ \mathbf{z}_s \odot \mathbf{z}_{m_i}, |\mathbf{z}_s - \mathbf{z}_{m_i}|,\ b_i]. \tag{7}$$

An absolute MLP produces a coarse score $c_i = f_{\mathrm{abs}}(\mathbf{x}_i) + \alpha b_i$. For the coarse top-$K_r$ candidates, let $u_i = W_m \mathbf{z}_{m_i}$ and $v = W_s \mathbf{z}_s$. The relation vector ρ_{ij} contains the spectrum condition, the signed difference $u_i - u_j$, its absolute value, the base-score difference $b_i - b_j$, and the cosine similarity between u_i and u_j. A shared MLP defines an ordered preference:

$$p_{ij} = g(\rho_{ij}) - g(\rho_{ji}), \qquad p_{ij} = -p_{ji}. \tag{8}$$

The relation weights are spectrum-conditioned and normalized over valid non-self candidates. With $w_{ij} \geq 0$, the final score is

$$q_i = c_i + \beta \sum_{j \neq i} w_{ij} p_{ij}. \tag{9}$$

The relation contribution is scattered back to the full candidate pool after coarse top-K_r selection. All operations are shared across candidates and use candidate-attached masks, so jointly permuting a list and its attached features permutes the output scores in the same way. Equation 8 gives pairwise antisymmetry independently of the learned parameter values. The model uses no raw peaks, SMILES tokens, precursor mass, formula, or acquisition metadata, and no rank embedding is present in its parameter path.

The capacity-matched pointwise control uses the same absolute branch and removes only the relative aggregation. At hidden width 512, the relative model has 1,485,189 trainable parameters and the no-rank pointwise control has 1,478,690 (0.44% difference). The generic candidate-set Transformer below is retained as a secondary legacy comparison, not as the proposed mechanism.

4.5 Legacy Candidate-Set Transformer Baseline

For comparability with the earlier audit, we retain the former rank-aware candidate-set Transformer and its pointwise control as a secondary baseline. They are not the formal model of this revision; their training-only forcing and rank shortcut are reported as historical evidence and are excluded from the new relative-model claims.

For candidate m_i with base rank r_i, we construct

$$\mathbf{x}_i = [\mathbf{z}_s,\ \mathbf{z}_{m_i},\ \mathbf{z}_s \odot \mathbf{z}_{m_i},\ |\mathbf{z}_s - \mathbf{z}_{m_i}|,\ b_i,\ \mathbf{e}(\bar{r}_i)], \tag{10}$$

where $\mathbf{e}(\bar{r}_i)$ is a learned rank embedding. The direct embeddings retain modality-specific evidence, while product and absolute-difference terms make symmetric matching cues explicit. An MLP projects each vector:

$$\mathbf{h}_i = \mathrm{MLP}(\mathbf{x}_i). \tag{11}$$

Both variants receive only the frozen spectrum embedding, frozen candidate embeddings, cosine base scores, clipped base-rank embeddings, and the candidate padding mask. Raw peaks, SMILES tokens, precursor mass or formula, and acquisition metadata are not passed directly to the reranker. Mass and formula constraints affect candidate-pool construction only; no explicit candidate–candidate structural-similarity feature is provided.

The candidate-set Transformer uses a Transformer encoder without candidate positional encodings to exchange information across candidates [17]:

$$[\tilde{\mathbf{h}}_1, \ldots, \tilde{\mathbf{h}}_K] = \text{Transformer}([\mathbf{h}_1, \ldots, \mathbf{h}_K]). \tag{12}$$

Padding masks support variable-sized lists. At deterministic inference, jointly permuting the candidates and their attached embeddings, scores, and ranks permutes the output scores in the same way; the base rank is a candidate-attached retrieval prior, not a tensor-slot position encoding [13,14]. The pointwise control sets $\tilde{\mathbf{h}}_i = \mathbf{h}_i$, retaining the same features, score head, residual path, and training losses while removing learned cross-candidate interaction after the retrieval priors have been constructed. It is not parameter- or capacity-matched because it removes the two Transformer layers: the pointwise and candidate-set variants contain exactly 1,478,690 and 7,783,458 trainable reranker parameters, respectively. A score head predicts a correction Δ_i, and the final score is residual:

$$q_i = \alpha b_i + \Delta_i, \tag{13}$$

where α is initialized to 1.0 and learned without a sign constraint. The residual path provides a direct shortcut from the base score; it does not guarantee preservation of the base order. Moreover, because b_i also enters $\mathbf{x}_i$ and Δ_i can learn additional score-dependent effects, α alone is not a mechanistically interpretable measure of reliance on the base retriever.

4.6 Learning Objective

For the labeled queries $\mathcal{B}_L$ in a minibatch, let V_n be the valid candidate set and y_n the positive index. The primary objective averages listwise cross-entropy over labeled queries:

$$\mathcal{L}_{\text{list}} = -\frac{1}{|\mathcal{B}_L|} \sum_{n \in \mathcal{B}_L} \log \frac{\exp(q_{ny_n})}{\sum_{i \in V_n} \exp(q_{ni})}. \tag{14}$$

We also contrast each positive with every other valid candidate retrieved for that query. The implementation averages over all valid positive–negative pairs in the minibatch:

$$\mathcal{L}_{\text{pair}} = \frac{\sum_{n \in \mathcal{B}_L} \sum_{i \in V_n \setminus \{y_n\}} \log(1 + \exp(q_{ni} - q_{ny_n} + \gamma))}{\sum_{n \in \mathcal{B}_L} (|V_n| - 1)}. \tag{15}$$

The complete objective is

$$\mathcal{L} = \mathcal{L}_{\text{list}} + \lambda_{\text{pair}} \mathcal{L}_{\text{pair}} + \lambda_{\text{spec}} \mathcal{L}_{\text{spec}}. \tag{16}$$

For a cyclicly mismatched spectrum s' from the same minibatch, the spectrum-dependency term contrasts the score of the true candidate under the correct and mismatched query:

$$\mathcal{L}_{\text{spec}} = \text{softplus}\big(q(s', m^+) - q(s, m^+) + \delta\big)\,. \tag{17}$$

It is an in-batch dependency constraint, not a claim of causal identification.

The listwise term is a top-one likelihood: it separates the positive from the valid list but does not prescribe a total order among negatives. Both terms are invariant to adding the same query-specific constant to every q_i. Thus, the raw outputs are relative scores, not probabilities; probability calibration is neither performed nor evaluated.

For embedding width d, full-pool coarse scoring costs $O(Kd)$ per query and the relative branch costs $O(K_r^2 d)$ with chunked pair evaluation, rather than quadratic work over all 256 candidates. This is a structural complexity statement; measured latency, throughput, and peak memory are reported only if the fixed-hardware benchmark in the analysis artifacts has completed.

5 Experiments

5.1 Dataset and Candidate Pools

We use the published MassSpecGym split [3]: 194,119 training spectra, 19,429 validation spectra, and 17,556 test spectra. These counts total 231,104 spectra; the benchmark reports approximately 29,000 unique molecular structures. Its similarity-aware, structure-disjoint construction reduces close structural overlap between training and test molecules.

An exact audit of spectrum-record signatures (peak arrays, precursor m/z, adduct, instrument, and collision energy) found no target-molecule duplicates across splits, but six validation and three test signatures exactly matched training signatures. In all nine matched pairs, the target molecules differ and each evaluation query is marked `simulation_challenge=true`. We report these cases as cross-split spectrum-record overlap. In the overlap-clean alignment-seed-42 pipeline used as our canonical configuration, all validation-dependent model selection excludes the six affected validation spectra, leaving 19,423 validation queries for the rerankers; alignment validation keys decrease from 3,386 to 3,384 molecular entries. We retain and evaluate all 17,556 published test queries.

MassSpecGym supplies mass- and formula-conditioned candidate settings. The mass pool uses a precursor-mass tolerance; the formula pool assumes a known formula and is not unrestricted annotation. Frozen pools contain at most 256 candidates. Before truncation, their test-query means are 253.88 and 165.75, respectively; 1 mass and 3,159 formula pools contain fewer than 40 candidates. After top-40 truncation, the means are 40.00 and 35.73.

Our primary evaluation follows the MassSpecGym 1.3.1 retrieval JSON order and the reference two-dimensional InChIKey identity rule. In official code commit `4f501b3`,[1] the reference loader instead defaults to two-dimensional InChIKey equivalence and may label multiple positives. The tracked full-pool identity audit found 17,556/17,556 positive queries and zero multiple-positive queries or identity collisions in both pools. Consequently, exact-target labels are equivalent to the reference identity labels on these saved candidate lists. This is an official-compatible candidate/identity evaluation on saved embeddings, not a fresh official-loader re-encoding or alignment retraining. The multiple-positive labels used during alignment are still cached raw exact-SMILES labels, not two-dimensional molecular-identity labels; this distinction is retained as a training-provenance limitation. The graph features do not explicitly encode atom chirality or bond stereochemistry, so a stereo variant could be treated differently by an identity rule outside this audited set.

The locally cached candidate sets exactly match the MassSpecGym 1.3.1 retrieval JSON sets, although their ordering differs (mass: 0/17,556 exact ordered-list matches; formula: 231/17,556). We therefore reconstructed the official JSON order, recomputed cosine/base ranks from the saved alignment embeddings, and evaluated all relative and pointwise reranker seeds 42–44. The official-order evaluation uses saved embeddings and does not rerun the full loader's transforms or retrain alignment; the resulting three-seed table is the primary result, while local exact-SMILES values are a sensitivity view.

5.2 Implementation Details

The aligned model outputs 512-dimensional embeddings. Its spectrum Transformer has four layers, 16 heads, and width 512; the molecular GINE has four layers, hidden width 128, and dropout 0.2. No published SpecEmbedding checkpoint is loaded, so the stage that would freeze a pretrained spectrum tower is skipped; both towers are optimized end to end on the MassSpecGym training split. Alignment uses AdamW with batch size 128, learning rate and weight decay 10^{-4}, cosine scheduling, gradient-norm clipping at 1, at most 100 epochs, and patience 5. Each epoch makes ten draws per molecular label, randomly selecting an associated spectrum. Spectrum and graph augmentation are independently enabled with probability 0.5. The former removes a floor-rounded 20% of fragment peaks below normalized intensity 0.3 and jitters intensities by $\pm 15\%$; the latter applies 10% node and 10% directed edge-entry dropout. Validation disables augmentation but retains replicate-spectrum sampling under the fixed random stream. The historical audit designates the overlap-clean project-default seed-42 run and its caches as canonical; seed 42 was frozen before its 43/44 sensitivity runs and was not selected by test performance. The new no-forcing pilot instead uses the available pre-overlap-clean alignment checkpoint `d4c1f70` and is reported separately, with no claim of overlap-clean or alignment-seed stability. It does not update the historical canonical artifact manifest.

The proposed reranker has hidden width 512, relation width 32, pair chunks of 16, and dropout 0.1. It scores the full candidate pool at $K = 256$ and applies

[1] MassSpecGym reference loader and label transform, accessed August 1, 2026.

the relative branch to its coarse top-$K_r = 40$. We use AdamW with learning rate 10^{-4}, batch size 16, at most 30 epochs, and early stopping on validation MRR with patience 5. The pairwise and spectrum-dependency weights and margins are $\lambda_{\text{pair}} = 0.2$, $\lambda_{\text{spec}} = 0.1$, $\gamma = 0.1$, and $\delta = 0.1$. The capacity-matched pointwise control uses the same absolute branch and optimization but no relation aggregation. The legacy candidate-set Transformer is trained only as a secondary historical comparison and retains its documented rank-aware configuration.

For each reported experiment, alignment checkpoints and reranker training seeds are listed with the result artifact. A reranker training seed controls initialization, data-loader order, candidate shuffling, and dropout; it is not an independent alignment replicate. Training uses only queries whose positive is naturally present in the full candidate pool. Queries without such a label are excluded from the reranker loss but retained in coverage statistics. No positive is inserted into any cache. All models are selected without test-set access and evaluated on one NVIDIA RTX 4090 when the corresponding run is available; CPU smoke tests are not presented as scientific results.

5.3 Code, Data, and Artifact Availability

The anonymous supplement provides source inspection, locked environment files, automated tests, and a tiny synthetic CPU smoke test. It does not provide the MassSpecGym spectra, candidate pools, trained alignment or reranker checkpoints, reranker caches, logs, or private analysis artifacts; the smoke test therefore does not reproduce the scientific metrics. The two pilot pools use separate mass and formula reranker trainings. The pilot training implementation is recorded at `488a04a`, the query-level evaluation utility at `8ce2655`, and the official-order evaluator in `official_protocol_eval.py`; parameter hashes are recorded in the internal pilot manifest alongside cache and checkpoint fingerprints. The frozen candidate artifacts and their provenance are recorded in the internal pilot manifest (not included in the anonymous submission); the release manifest covers only manuscript PDFs. The earlier rank-aware Transformer audit remains traceable through its own historical artifact records and is not merged with the pilot provenance. The disclosed graph features are atom symbol, degree, total hydrogen count, aromaticity, ring-size bin, formal charge; bond type, conjugation, and ring membership; and $\log(1+n_{\text{atoms}})$, $\log(1+n_{\text{bonds}})$ graph-size features. The alignment projection is 512-dimensional with temperature $\tau = 0.07$; AdamW uses weight decay 10^{-4} and gradient clipping 1.0; spectrum caches are FP32 and molecule caches FP16. The real-data entry point is `run_rerank_pipeline.py`, subject to access to the excluded data and artifacts.

5.4 Baselines and Metrics

The controlled comparisons are the same aligned cosine retriever, the capacity-matched no-rank pointwise coarse scorer, the proposed relative reranker, and the legacy candidate-set Transformer. The latter is retained as a mechanism baseline rather than the formal method. We retain JESTR and GLMR as reported-only

Table 2. Official-compatible no-forcing full-pool pilot. Values are percentages; learned rows are means ± sample SD over reranker training seeds 42–44.

Pool	Method	R@1	R@5	R@20	R@40	MRR
Mass	SpecMolAlign (full pool)	43.78	61.10	75.55	82.15	52.19
	Capacity-matched pointwise	51.37±0.54	67.34±0.39	80.71±0.55	86.03±0.61	59.03±0.44
	Relative candidate model	50.98±0.87	67.06±0.67	80.49±0.55	86.13±0.70	58.71±0.76
Formula	SpecMolAlign (full pool)	58.49	71.50	81.81	87.30	64.79
	Capacity-matched pointwise	66.65±1.30	76.76±0.87	85.33±0.42	89.43±0.29	71.66±1.04
	Relative candidate model	66.32±0.46	76.29±0.23	85.08±0.07	89.31±0.06	71.29±0.33

mechanism background from the GLMR paper [20]; no external numeric values are shown. They are not a controlled comparison: JESTR and GLMR were not independently reproduced in our codebase, and their retrievers, candidate handling, identity/canonicalization rules, and MCES implementations are not matched to our local exact-target-SMILES single-positive protocol.

We report Recall@1, Recall@5, Recall@20, Recall@40, and mean reciprocal rank (MRR) over all 17,556 test queries. The main table uses the official retrieval JSON order and the audited 2D-identity-equivalent labels on saved embeddings. The pilot uses three reranker training seeds (42–44) on the same pre-overlap-clean alignment checkpoint and 5,000 training queries; means and sample SDs are seed dispersion, not independent alignment replicates. The query-level seed-42 paired bootstrap uses the saved per-query predictions and 2,000 deterministic resamples under the local sensitivity protocol. The new pilot does not recompute MCES; the earlier thresholded myopic-MCES audit remains a separate single-checkpoint artifact.

5.5 Main Results

Both learned models improve on their full-pool base in both pools. Pointwise is slightly higher than relative at R@1 and MRR in the official-compatible three-seed aggregate (mass: 51.37 vs. 50.98 and 0.5903 vs. 0.5871; formula: 66.65 vs. 66.32 and 0.7166 vs. 0.7129). Evaluation-only K sensitivity on the same 256-candidate cache gives mass relative/pointwise R@1-at-K of 80.49/80.71 at $K = 20$, 86.13/86.03 at $K = 40$, and 93.45/93.38 at $K = 80$; these are not K-specific training runs. The seed-42 paired bootstrap gives a relative-minus-base R@1 difference of 6.30 percentage points (95% CI 5.69–6.93) and MRR difference 0.0576 (0.0530–0.0625) for mass. New pilot runs did not compute MCES; the historical single-checkpoint MCES audit remains separate.

As a local exact-target-SMILES sensitivity check, the corresponding three-seed means differ from the official-compatible table only below the displayed precision for these caches. This agreement is conditional on the audited candidate sets containing no 2D-identity collisions; it is not evidence that a fresh loader reconstruction or a different candidate source would behave the same way.

Information-source and mechanism audit. On the same seed-42 pilot, a base-score-only control reaches 43.78%/0.5219 (mass) and 58.49%/0.6479 (formula) for R@1/MRR, i.e., no improvement over the base. An embedding-only absolute scorer reaches 50.07%/0.5808 and 66.13%/0.7110, while adding the base-score pathway without the relative module gives 50.07%/0.5798 and 66.28%/0.7141. The full relative model gives 50.08%/0.5795 and 66.80%/0.7161. Candidate-only controls (45.00%/0.5245 and 52.73%/0.5960) are weaker than spectrum-conditioned variants. Removing molecular relations, spectrum conditioning, or antisymmetric pairing changes the two pools in different directions; these single-seed controls therefore support an information-source finding, not a causal component ranking.

JESTR and GLMR remain reported-only, not reproduced, and not directly comparable background methods. We retain their mechanism descriptions but do not reproduce their numeric table or infer a relative ranking.

Difficulty strata. In the seed-42 local query analysis, relative gains concentrate among queries that the base ranker leaves unresolved. For base ranks 2–5/6–20, relative R@1 gains over base are +42.17/+23.02 percentage points for mass and +52.72/+30.06 for formula; when the base top-1 Morgan similarity is below 0.25, the gains are +24.59/+35.92 points. Base-rank-1 queries and top-1 similarity at least 0.5 instead decline (mass -13.28/-11.86; formula -5.41/-3.65). These are descriptive seed-42 strata under the local protocol, not causal or significance evidence for the relative module.

Historical audit boundary. Earlier rank-aware Transformer results used the former top-40 cache and training-only positive replacement. They remain available in the internal analysis artifacts for traceability, but are not measurements of the present rank-free, no-forcing method and are not used for its claims. In particular, their multi-seed and cross-alignment summaries must not be read as evidence for the relative module introduced here.

6 Discussion

The new full-pool pilot separates candidate coverage from list ordering. Its full-pool upper bound is 100% by construction of the saved cache, whereas evaluation-time top-40 truncation gives a mass upper bound of 82.15%. Across three reranker seeds, relative and pointwise both improve the base; pointwise is slightly stronger at R@1/MRR, while relative is not uniformly better at any cut-off. A fixed RTX 4090 forward-only audit (batch 64, 2,048 queries, data loading excluded) decomposes mass relative into 0.261 ms/query full-pool coarse scoring plus 0.111 ms/query top-40 relation scoring (0.372 ms/query, 2,685 q/s, 302 MB peak), compared with 0.030 ms/query, 33,450 q/s and 244 MB for pointwise. Formula relative is 0.026+0.050=0.076 ms/query (13,235 q/s, 302 MB), compared with 0.026 ms/query, 38,492 q/s and 244 MB. These are implementation audits rather than end-to-end efficiency claims.

The mass spectrum-shuffle audit reduces the relative model to 45.05% Recall@1 and 0.5358 MRR, still above its base but well below the correctly paired relative result. The drop is evidence that the score is not completely spectrum-independent, but the residual performance under a shuffled spectrum also shows that candidate and base-score pathways contribute materially. We therefore do not attribute the pointwise-to-relative gap to a causal benefit of candidate relations. The historical Transformer results below are retained only as an audit of the previous design and are not merged with this pilot.

The official-compatible evaluation uses the published candidate JSON and saved alignment embeddings, but does not re-encode spectra/molecules through a fresh official loader or retrain the first stage. It therefore does not establish end-to-end equivalence for a separately reconstructed loader, a different candidate source, or an external method. JESTR/GLMR remain reported-only and were not reproduced.

7 Limitations and Responsible Evaluation

The primary new evidence uses 5,000 of 194,119 training queries for three epochs and three reranker training seeds on one alignment checkpoint. This is not alignment-level or end-to-end seed stability. The official-compatible results use saved embeddings and a full-pool 2D-InChIKey identity audit, but not fresh official-loader re-encoding; external baselines and candidate-aware alignment retraining remain outside scope. The query-level bootstrap is descriptive and does not replace independent retraining.

The no-forcing cache retains only candidates supplied by the MassSpecGym files. The reranker cannot recover a target absent from an evaluation cutoff, and it cannot propose a structure outside the closed candidate pool. Candidate relations are computed only on the coarse top-40, so claims do not extend to a different cutoff or to an unrestricted database. The spectrum-shuffle result is an in-batch dependency audit, not a causal or invariance guarantee.

The anonymous supplement provides code inspection, locked environments, tests, and a synthetic smoke test. It does not include the real data, candidate files, caches, checkpoints, logs, or pilot-result artifacts. Matched external baselines and candidate-aware alignment retraining are outside scope. The reported latency and memory numbers are fixed-hardware forward audits, not end-to-end throughput or deployment guarantees.

The spectrum tower does not receive adduct, instrument, or collision-energy metadata, and formula conditioning presumes a known formula. Transfer to other acquisition conditions, candidate constructions, identity rules, benchmarks, or larger lists is untested. Historical forced-positive Transformer results are kept separate and do not repair the new model's evidence boundary.

8 Conclusion

We introduced SpecEmbedding-Rerank, a rank-free, no-forcing second-stage reranker that combines full-pool coarse scoring with spectrum-conditioned relative candidate preferences. Across three reranker training seeds under the

MassSpecGym 1.3.1- compatible candidate-order and two-dimensional-identity protocol on saved embeddings, both the relative model and its capacity-matched pointwise control improved the full-pool base in mass- and formula-conditioned retrieval. Pointwise was slightly stronger at the main R@1/MRR summaries, and the spectrum-shuffle audit retained a substantial candidate/base-score contribution. The evidence therefore supports a carefully scoped supervised reranking framework, but does not establish an independent or generally robust benefit of the relative module. Fresh official-loader re-encoding, external-baseline comparison, and end-to-end deployment efficiency remain unclaimed.

References

1. Bach, E., Schymanski, E.L., Rousu, J.: Joint structural annotation of small molecules using liquid chromatography retention order and tandem mass spectrometry data. Nature Machine Intelligence **4**(12), 1224–1237 (2022). https://doi.org/10.1038/s42256-022-00577-2
2. Burges, C., Shaked, T., Renshaw, E., Lazier, A., Deeds, M., Hamilton, N., Hullender, G.: Learning to rank using gradient descent. In: Proceedings of the 22nd International Conference on Machine Learning. pp. 89–96 (2005). https://doi.org/10.1145/1102351.1102363
3. Bushuiev, R., Bushuiev, A., de Jonge, N.F., Young, A., Kretschmer, F., Samusevich, R., Heirman, J., Wang, F., Zhang, L., Dührkop, K., Ludwig, M., Haupt, N.A., Kalia, A., Brungs, C., Schmid, R., Greiner, R., Wang, B., Wishart, D.S., Liu, L.P., Rousu, J., Bittremieux, W., Röst, H., Mak, T.D., Hassoun, S., Huber, F., van der Hooft, J.J.J., Stravs, M.A., Böcker, S., Sivic, J., Pluskal, T.: MassSpecGym: A benchmark for the discovery and identification of molecules. In: Advances in Neural Information Processing Systems. vol. 37, pp. 110010–110027 (2024). https://doi.org/10.52202/079017-3491
4. Cao, Z., Qin, T., Liu, T.Y., Tsai, M.F., Li, H.: Learning to rank: From pairwise approach to listwise approach. In: Proceedings of the 24th International Conference on Machine Learning. pp. 129–136 (2007). https://doi.org/10.1145/1273496.1273513
5. Chen, L., Xia, B., Wang, Y., Huang, X., Gu, Y., Wu, W., Zhou, Y.: CMSSP: A contrastive mass spectra-structure pretraining model for metabolite identification. Analytical Chemistry **96**(42), 16871–16881 (2024). https://doi.org/10.1021/acs.analchem.4c03724
6. Chen, Y.Z., Hassoun, S.: Learning from all views: A multiview contrastive framework for metabolite annotation. Analytical Chemistry **98**(9), 6598–6606 (2026). https://doi.org/10.1021/acs.analchem.5c05675
7. Dührkop, K., Shen, H., Meusel, M., Rousu, J., Böcker, S.: Searching molecular structure databases with tandem mass spectra using CSI:FingerID. Proceedings of the National Academy of Sciences **112**(41), 12580–12585 (2015). https://doi.org/10.1073/pnas.1509788112
8. Gerlich, M., Neumann, S.: MetFusion: Integration of compound identification strategies. Journal of Mass Spectrometry **48**(3), 291–298 (2013). https://doi.org/10.1002/jms.3123
9. Goldman, S., Wohlwend, J., Stražar, M., Haroush, G., Xavier, R.J., Coley, C.W.: Annotating metabolite mass spectra with domain-inspired chemical formula trans-

formers. Nature Machine Intelligence **5**(9), 965–979 (2023). https://doi.org/10.1038/s42256-023-00708-3

10. Hu, W., Liu, B., Gomes, J., Zitnik, M., Liang, P., Pande, V., Leskovec, J.: Strategies for pre-training graph neural networks. In: International Conference on Learning Representations (2020), https://openreview.net/forum?id=HJlWWJSFDH
11. de Jonge, N.F., Louwen, J.J.R., Chekmeneva, E., Camuzeaux, S., Vermeir, F.J., Jansen, R.S., Huber, F., van der Hooft, J.J.J.: MS2Query: Reliable and scalable MS2 mass spectra-based analogue search. Nature Communications **14**(1), 1752 (2023). https://doi.org/10.1038/s41467-023-37446-4
12. Kalia, A., Chen, Y.Z., Krishnan, D., Hassoun, S.: JESTR: Joint embedding space technique for ranking candidate molecules for the annotation of untargeted metabolomics data. Bioinformatics **41**(7), btaf354 (2025). https://doi.org/10.1093/bioinformatics/btaf354
13. Lee, J., Lee, Y., Kim, J., Kosiorek, A., Choi, S., Teh, Y.W.: Set transformer: A framework for attention-based permutation-invariant neural networks. In: Proceedings of the 36th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 97, pp. 3744–3753. PMLR (2019), https://proceedings.mlr.press/v97/lee19d.html
14. Pang, L., Xu, J., Ai, Q., Lan, Y., Cheng, X., Wen, J.: SetRank: Learning a permutation-invariant ranking model for information retrieval. In: Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval. pp. 499–508 (2020). https://doi.org/10.1145/3397271.3401104
15. Stravs, M.A., Dührkop, K., Böcker, S., Zamboni, N.: MSNovelist: De novo structure generation from mass spectra. Nature Methods **19**(7), 865–870 (2022). https://doi.org/10.1038/s41592-022-01486-3
16. Tian, Y., Krishnan, D., Isola, P.: Contrastive multiview coding. In: Computer Vision – ECCV 2020. Lecture Notes in Computer Science, vol. 12356, pp. 776–794. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-58621-8_45
17. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I.: Attention is all you need. In: Advances in Neural Information Processing Systems. vol. 30, pp. 5998–6008 (2017), https://proceedings.neurips.cc/paper/7181-attention-is-all-you-need
18. Xiong, P., Xu, H., Zheng, H.: Supervised contrastive learning leads to more reasonable spectral embeddings. Analytical Chemistry **97**(37), 20137–20146 (2025). https://doi.org/10.1021/acs.analchem.5c02655
19. Young, A., Röst, H., Wang, B.: Tandem mass spectrum prediction for small molecules using graph transformers. Nature Machine Intelligence **6**(4), 404–416 (2024). https://doi.org/10.1038/s42256-024-00816-8
20. Zhang, Y., Ding, K., Wu, Y., Zhuang, X., Yang, Y., Zhang, Q., Chen, H.: Breaking the modality barrier: Generative modeling for accurate molecule retrieval from mass spectra. Proceedings of the AAAI Conference on Artificial Intelligence **40**(2), 1561–1569 (2026). https://doi.org/10.1609/aaai.v40i2.37132